INCH-POUND

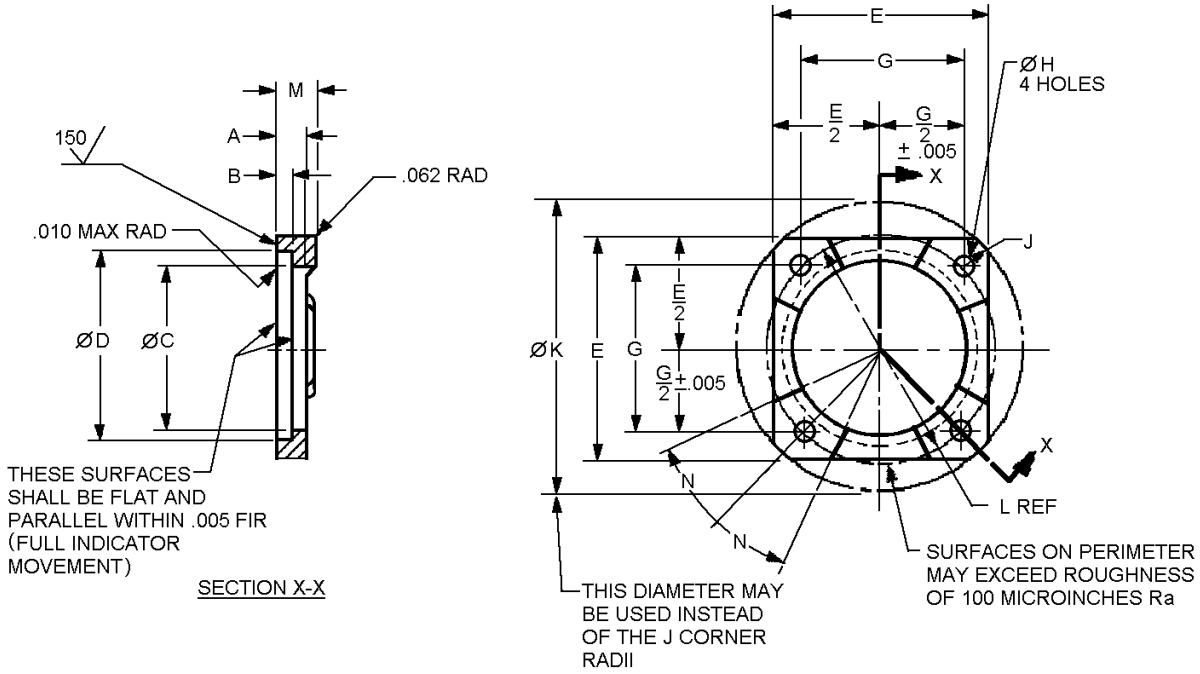
MS20756E
w/AMENDMENT 1
26 October 2010
SUPERSEDING
MS27056E
17 November 2009

DETAIL SPECIFICATION SHEET

FLANGE, SWIVEL, RETAINING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and SAE-AS4875.




| Inches | mm |
|---|---|
| .005 | 0.13 |
| .010 | 0.25 |
| .062 | 1.57 |

FIGURE 1. Swivel fitting flange type I rectangle size 8 through 32.



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| Size designator | Tubing OD and hose ID inches (mm) | A +.015 -.005 (+0.38 -0.13) inches (mm) | B +.005/-.000 (+0.13 -0.000) inches (mm) | C dia (see note 4) +.010/-.000 (+0.25 -0.000) inches (mm) | D dia (see note 4) +.005 (+0.13 -.000) inches (mm) | E inches (mm) | |
|---|---|---|---|---|---|---|---|
| 8 | .500 (12.70) | .234 (5.94) | .136 (3.45) | .750 (19.05) | .885 (22.48) | 1.390 (35.31) | ±.016 (0.41) |
| 10 | .625 (15.88) | | | .880 (22.35) | 1.010 (25.65) | 1.468 (37.29) | |
| 12 | .750 (19.05) | | | 1.125 (28.58) | 1.260 (32.00) | 1.594 (40.49) | |
| 16 | 1.000 (25.40) | | | 1.375 (34.93) | 1.510 (38.35) | 1.750 (44.45) | |
| 20 | 1.250 (31.75) | | | 1.688 (42.88) | 1.854 (47.09) | 2.188 (55.58) | ±.020 (0.51) |
| 24 | 1.500 (38.10) | .297 (7.54) | .168 (4.27) | 1.938 (49.23) | 2.135 (54.23) | 2.375 (60.33) | |
| 32 | 2.000 (50.80) | | | 2.562 (65.07) | 2.760 (70.10) | 3.000 (76.20) | |

FIGURE 1. Swivel fitting flange type I rectangle size 8 through 32 – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:24Z
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| Size number | G ±.005 (±0.13) inches (mm) | H dia +.010/-.000 (+0.25 -0.000) inches (mm) | J radius ±.016 (±0.41) inches (mm) | K dia ±.015 (±0.38) inches (mm) | L dia inches (mm) | M ±.016 (±0.41) inches (mm) | N |
|---|---|---|---|---|---|---|---|
| 8 | .950 (24.13) | .205 (5.21) | .219 (5.56) | 1.782 (45.26) | 1.344 (34.14) | .296 (7.52) | 20° |
| 10 | 1.038 (26.37) | | | 1.906 (48.41) | 1.468 (37.29) | | |
| 12 | 1.156 (29.36) | | | 2.094 (53.19) | 1.635 (41.53) | | |
| 16 | 1.312 (33.32) | | | 2.312 (58.72) | 1.855 (47.12) | | 18° |
| 20 | 1.656 (42.06) | .266 (6.76) | .266 (6.76) | 2.875 (73.03) | 2.342 (59.49) | | 16° |
| 24 | 1.812 (46.02) | | .281 (7.14) | 3.094 (78.59) | 2.562 (65.07) | .359 (9.12) | |
| 32 | 2.375 (60.33) | .328 (8.33) | .312 (7.92) | 3.953 (100.41) | 3.359 (85.32) | | 12° |

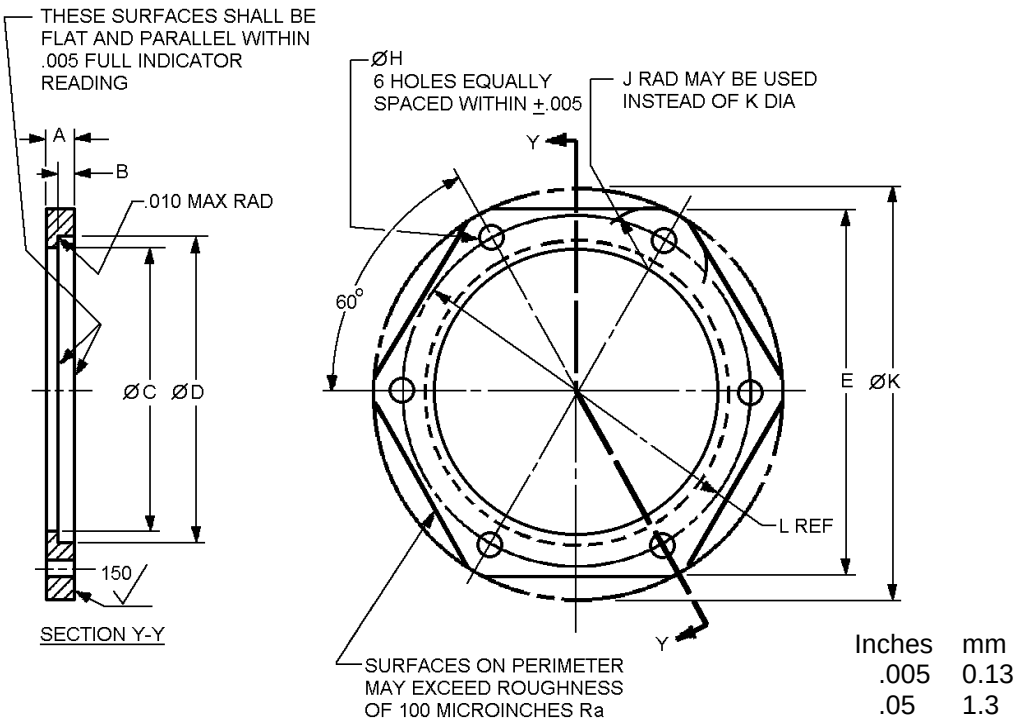
| Size number | Weight max (estimated) lbs (gram) | | |
|---|---|---|---|
| | Steel | Al alloy | Titanium |
| 8 | .03 (14) | .01 (5) | .02 (9) |
| 10 | .05 (23) | .02 (9) | .03 (14) |
| 12 | .08 (36) | .03 (14) | .04 (18) |
| 16 | | | |
| 20 | .14 (64) | .05 (23) | .07 (32) |
| 24 | .17 (77) | .06 (27) | .09 (41) |
| 32 | .25 (113) | .09 (41) | .13 (59) |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise, tolerances, ±.005 inch (0.13 mm) angles ±.5°.
4. Diameters C and D shall be concentric within .010 inch (0.25 mm) full indicator reading.
5. Unless otherwise specified, the surface roughness shall not exceed 100 microinches (25.4 µm) Ra in accordance with ASME B46.1.
6. Break all sharp edges; remove all loose or hanging burrs and slivers that may become dislodged.
7. For tube sizes -8 through -16, maximum operating temperature 275°F (135°C), maximum pressure 1500 psi (10 MPa).
8. For tube sizes -20 though -48, maximum operating temperature 275°F (135°C), maximum pressure 500 psi (3.4 MPa).

FIGURE 1. Swivel fitting flange type I rectangle size 8 through 32 – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:24Z
Check the source to verify that this is the current version before use.




| Size designator | Tubing OD and hose ID inches (mm) | A +.015 -.005 (+0.38 -0.13) inches (mm) | B +.005/-.000 (+0.13 -0.000) inches (mm) | C dia (see note 4) +.010/-.000 (+0.25 -0.000) inches (mm) | D dia (see note 4) +.005 (+0.13 -.000) inches (mm) | E inches (mm) | |
|---|---|---|---|---|---|---|---|
| 8 | .500 (12.70) | .234 (5.94) | .136 (3.45) | .750 (19.05) | .885 (22.48) | 1.390 (35.31) | ±.016 (0.41) |
| 10 | .625 (15.88) | | | .880 (22.35) | 1.010 (25.65) | 1.468 (37.29) | |
| 12 | .750 (19.05) | | | 1.125 (28.58) | 1.260 (32.00) | 1.594 (40.49) | |
| 16 | 1.000 (25.40) | | | 1.375 (34.93) | 1.510 (38.35) | 1.750 (44.45) | |
| 20 | 1.250 (31.75) | | | 1.688 (42.88) | 1.854 (47.09) | 2.188 (55.58) | ±.020 (0.51) |
| 24 | 1.500 (38.10) | .297 (7.54) | .168 (4.27) | 1.938 (49.23) | 2.135 (54.23) | 2.375 (60.33) | |
| 32 | 2.000 (50.80) | | | 2.562 (65.07) | 2.760 (70.10) | 3.000 (76.20) | |
| 40 | 2.500 (63.50) | | | 3.062 (77.77) | 3.291 (83.59) | 4.000 (101.60) | |
| 48 | 3.000 (76.20) | | | 3.562 (90.47) | 3.791 (96.29) | 4.500 (114.30) | |

FIGURE 2. Swivel fitting flange type II hexagon size 8 through 48.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:24Z
Check the source to verify that this is the current version before use.

| Size number | G ±.005 (±0.13) inches (mm) | H dia +.010/-.000 (+0.25 -0.000) inches (mm) | J radius ±.016 (±0.41) inches (mm) | K dia ±.015 (±0.38) inches (mm) | L dia inches (mm) | M ±.016 (±0.41) inches (mm) | N |
|---|---|---|---|---|---|---|---|
| 8 | .950 (24.13) | .205 (5.21) | .219 (5.56) | 1.782 (45.26) | 1.344 (34.14) | .296 (7.52) | 20° |
| 10 | 1.038 (26.37) | | | 1.906 (48.41) | 1.468 (37.29) | | |
| 12 | 1.156 (29.36) | | | 2.094 (53.19) | 1.635 (41.53) | | |
| 16 | 1.312 (33.32) | | | 2.312 (58.72) | 1.855 (47.12) | | 18° |
| 20 | 1.656 (42.06) | .266 (6.76) | .266 (6.76) | 2.875 (73.03) | 2.342 (59.49) | | 16° |
| 24 | 1.812 (46.02) | | .281 (7.14) | 3.094 (78.59) | 2.562 (65.07) | .359 (9.12) | |
| 32 | 2.375 (60.33) | .328 (8.33) | .312 (7.92) | 3.953 (100.41) | 3.359 (85.32) | | 12° |
| 40 | - | | .375 (9.53) | 4.500 (114.30) | 3.812 (96.82) | - | - |
| 48 | - | | .625 (15.88) | 5.000 (127.00) | 4.312 (109.52) | - | - |

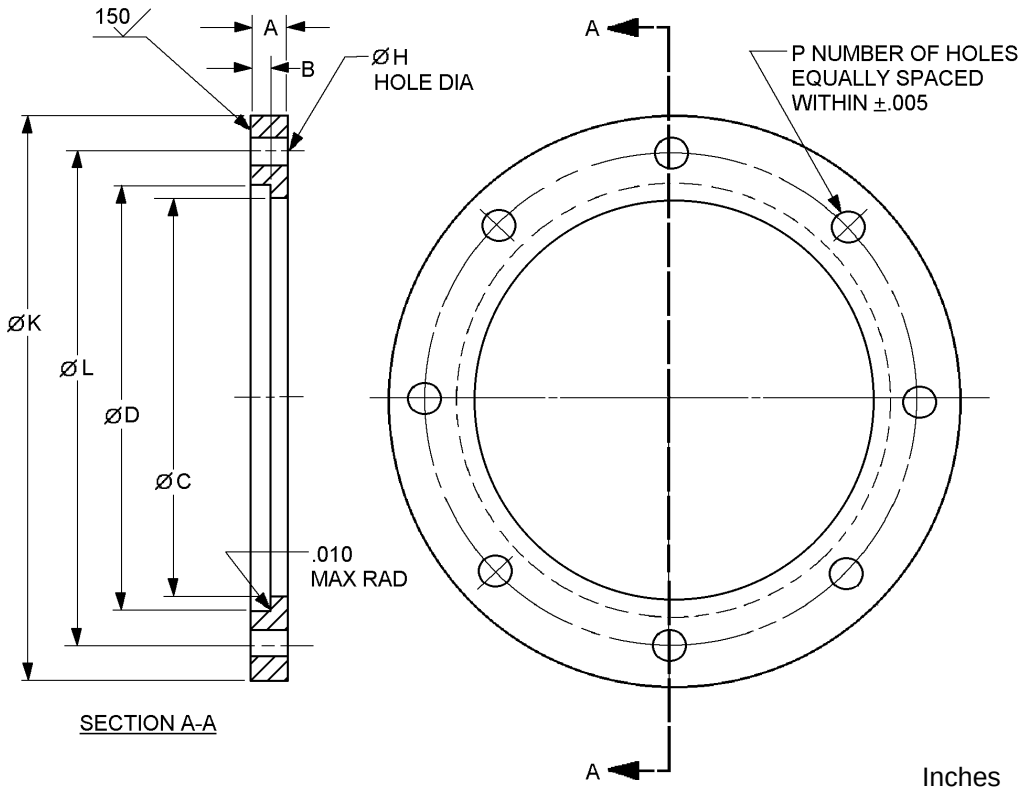
| Size number | Weight max (estimated) lbs (gram) | | |
|---|---|---|---|
| | Steel | Al alloy | Titanium |
| 8 | .03 (14) | .01 (5) | .02 (9) |
| 10 | .05 (23) | .02 (9) | .03 (14) |
| 12 | .08 (36) | .03 (14) | .04 (18) |
| 16 | | | |
| 20 | .14 (64) | .05 (23) | .07 (32) |
| 24 | .17 (77) | .06 (27) | .09 (41) |
| 32 | .25 (113) | .09 (41) | .13 (59) |
| 40 | .45 (204) | .16 (73) | .21 (95) |
| 48 | .53 (240) | .19 (86) | .24 (109) |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise, tolerances, ±.005 inch (0.13 mm) angles ±.5°.
4. Diameters C and D shall be concentric within .010 inch (0.25 mm) full indicator reading.
5. Unless otherwise specified, the surface roughness shall not exceed 100 microinches (25.4 µm) Ra in accordance with ASME B46.1.
6. Break all sharp edges; remove all loose or hanging burrs and slivers that may become dislodged.
7. For tube sizes -8 through -16, maximum operating temperature 275°F (135°C), maximum pressure 1500 psi (10 MPa).
8. For tube sizes -20 though -48, maximum operating temperature 275°F (135°C), maximum pressure 500 psi (3.4 MPa).

FIGURE 2. Swivel fitting flange type II hexagon size 8 through 48 - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:24Z
Check the source to verify that this is the current version before use.




| Inches | mm |
|---|---|
| .005 | 0.13 |
| .010 | 0.25 |

| Size number | A +.015/ -.005 (+0.38/ -(0.13) inches (mm) | B +.005/ -.000 (+0.13/ -.000) inches (mm) | C dia (see note 4) .031 (0.79) inches (mm) | D dia (see note 4) inches (mm) | | H dia +.010/ -.000 (+0.25/ -0.00) inches (mm) | K dia ±.015 (±0.38) inches (mm) |
|---|---|---|---|---|---|---|---|
| 56 | .312 (7.92) | .168 (4.27) | 3.968 (100.79) | 4.234 (107.54) | +.010/ -.000 (+0.25/ -0.000) | .328 (8.33) | 5.562 (141.27) |
| 64 | | | 4.531 (115.09) | 4.797 (121.84) | | | 6.125 (155.58) |
| 80 | .375 (9.53) | .188 (4.78) | 5.531 (140.49) | 5.797 (147.24) | | | 7.125 (180.98) |
| 96 | | | 6.531 (165.89) | 6.797 (172.64) | +.020/ -.000 (0.51/ -.000) | | 8.125 (206.38) |
| 112 | .437 (11.10) | .200 (5.08) | 7.594 (192.89) | 7.860 (199.64) | | | 9.188 (233.38) |

FIGURE 3. Swivel fitting flange type II hexagon size 56 through 112.



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| Size number | L dia Ref inches (mm) | P number of holes |
|---|---|---|
| 56 | 4.875 (123.83) | 8 |
| 64 | 5.438 (138.13) | |
| 80 | 6.438 (163.53) | 10 |
| 96 | 7.438 (188.93) | 12 |
| 112 | 8.500 (215.90) | 14 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise, tolerances, ±.005 inch (0.13 mm), angles ±.5°.
4. Diameters C and D shall be concentric within .010 inch (0.25 mm) full indicator reading.
5. Unless otherwise specified, the surface roughness shall not exceed 100 microinches (2.54 µm) Ra in accordance with ASME B46.1.
6. Break all sharp edges; remove all loose or hanging burrs and slivers that may become dislodged.
7. Maximum operating temperature 275°F (135°C), maximum pressure 500 psi (3.4 MPa).

FIGURE 3. Swivel fitting flange type II hexagon size 56 through 112 – Continued.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figures 1, 2, and 3.

Reduction by forging draft angle of 7° maximum is permissible.

Materials. Materials shall be in accordance with MIL-DTL-83798.

Finishes. Finishes for the flange shall be in accordance with MIL-DTL-83798.

Finish designators shall be as specified in table I.



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TABLE I. Flange finish code. 1/

| PIN code dash letter | Material | Plating finish |
|---|---|---|
| Blank | Steel | Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2. |
| | Aluminum | Anodize in accordance with MIL-A-8625, type II. 2/ |
| | Corrosion resistant steel | No additional finish. Passivation in accordance with SAE-AMS2700, type 6 or 7. |
| -A | Steel | Aluminum-nickel in accordance with ASTM F1136, grade 3, NC |
| -D | Aluminum | Anodize above with NAVAIR trivalent chromium pretreatment (TCP) in accordance with MIL-DTL-81706, type 2, class 1A. 2/ |
| -CN | Steel | Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2 with NAVAIR TCP in accordance with MIL-DTL-81706, type 2, class 1A. |
| -F | Steel | TCP in accordance with MIL-DTL-81706, type 2, class 1A. 2/ |
| -G | Steel | Zinc plating with colorless passivate in accordance with ASTM B633, type V, Fe/Zn 25. |
| -H | Steel | Zinc phosphate finish in accordance MIL-DTL-16232 type Z, class1. 3/ |
| -J | Steel | Zinc plating in accordance with ASTM B633; type II or III, Fe/Zn 5, or ASTM B695, type II, class 5. |
| -N | Steel | NAVAIR TCP in accordance with MIL-DTL-81706, type 2, class A. |
| -R | Steel | Zinc plating in accordance with ASTM B633; type VI, Fe/Zn 5. |
| -T | Titanium 4/ | Anodized in accordance with SAE-AMS2488, type 2 or fluoride phosphated in accordance with SAE-AMS2486. 5/ |
| -V | Steel | Zinc-nickel in accordance with SAE-AMS2417, type 1. |
| -Z | Zinc any type above | PIN code H, J, R, V. |
| -ZN | Zinc any type above | PIN code H, J, R, V with NAVAIR TCP in accordance with MIL-DTL-81706, type 2, class 1A. |

1/ Shall be capable of withstanding minimum of 96 hours salt spray.
2/ Die light blue.
3/ Hexavalent chromium free.
4/ Titanium shall not be used in oxygen or potable water systems.
5/ Color equivalent to numbers /36076, /36081, /36099, /36118, /36134, /36152, /36170, /36173, and /36176 in accordance with FED-STD-595.

Trivalent wrenchability. When the finish has been damaged due to poor wrenchability, the surface of the connector shall be touched up using the brush plating process below. The term “trivalent wrenchability” is used to evaluate the ability of the finish to withstand abrasion from an excessive amount of wrenching

a. Brush plating of hard chromium by electrodeposition shall be in accordance with SAE-AMS-2451/5.

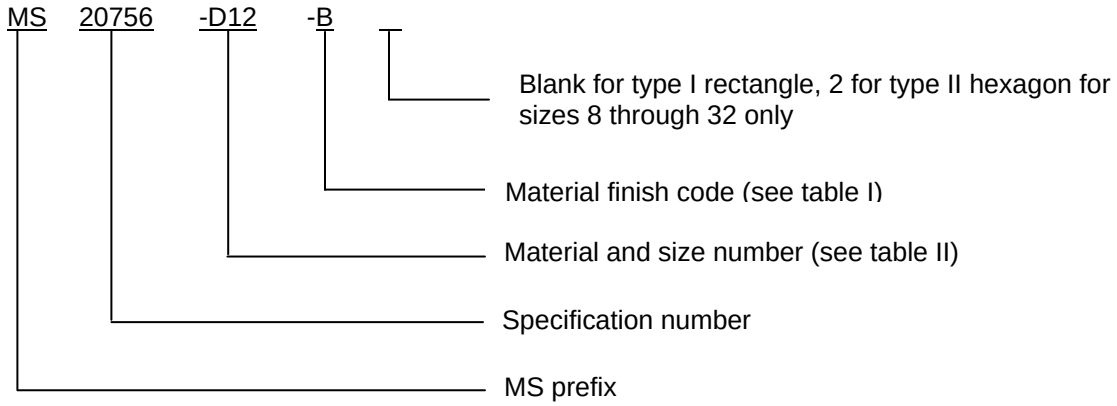
b. Brush plating of medium-hardness, low stress nickel by electrodeposition shall be in accordance with SAE-AMS-2451/9.

c. Brush plating of NAVAIR TCP shall be in accordance with MIL-DTL-81706, type 2, class A, material form 1 through 6, application method B. Example of a PIN: M817062A6B.



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Part or Identifying Number (PIN) example:




MS20756-D12-B indicates: Flange, .750 inch tube ID, aluminum anodized with NAVAIR TCP, type I rectangle.

MS20756-D12-B2 indicates: Flange, .750 inch tube ID, aluminum anodized with NAVAIR TCP, type II hexagon.

Material and size designators. Material and size designators shall be as specified in table II.

TABLE II. Material and size designators.

| Size dash number with letter material designator | | | |
|---|---|---|---|
| Steel | Aluminum alloy | CRES 1/ | Titanium 2/ |
| -8 | -D8 | -J8 | -T8 |
| -10 | -D10 | -J10 | -T10 |
| -12 | -D12 | -J12 | -T12 |
| -16 | -D16 | -J16 | -T16 |
| -20 | -D20 | -J20 | -T20 |
| -24 | -D24 | -J24 | -T24 |
| -32 | -D32 | -J32 | -T32 |
| -40 | -D40 | -J40 | -T40 |
| -48 | -D48 | -J48 | -T48 |
| -56 | -D56 | -J56 | -T56 |
| -64 | -D64 | -J64 | -T64 |
| -80 | -D80 | -J80 | -T80 |
| -96 | -D96 | -J96 | -T96 |
| -112 | -D112 | -J112 | -T112 |

1/ Corrosion resistant steel (CRES).
2/ Titanium shall not be used in potable water or oxygen systems.

Marking. The complete PIN shall be permanently marked.



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Guidance on use of alternative parts with less hazardous or nonhazardous materials. This specification provides for a number of alternative plating materials via the PIN. Users should select the PIN with the least hazardous material that meets the form, fit and function requirements of their application.

For nominal use on fuel and oil systems.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to SAE-AS4875, this document references the following:

| | |
|---|---|
| MIL-A-8625 | ASME B46.1 |
| MIL-DTL-16232 | ASTM B633 |
| MIL-DTL-81706 | ASTM B695 |
| MIL-DTL-83798 | ASTM F1136 |
| FED-STD-595/36076 | SAE-AMS-C-81562 |
| FED-STD-595/36081 | SAE-AMS-QQ-P-416 |
| FED-STD-595/36099 | SAE-AMS2700 |
| FED-STD-595/36118 | SAE-AMS2417 |
| FED-STD-595/36134 | SAE-AMS-2451/5 |
| FED-STD-595/36152 | SAE-AMS-2451/9 |
| FED-STD-595/36170 | SAE-AMS2486 |
| FED-STD-595/36173 | SAE-AMS2488 |
| FED-STD-595/36176 | |

CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2010-063)

Review activities:
Army - AT
Navy – SA
Air Force - 85

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